Failure Analysis Report

FAIL-0164 — Timeout

1. Failure Summary

Failure ID	FAIL-0164
Failure Type	Timeout
Severity	Trivial
Component	Sensor Manager
Application	APP-012 — SecureAuthentication
Build Version	4.3.2
Timestamp	2024-12-03 21:01:08
Environment	ENV-009 — Low Network Env
OS Version	Android 14
Network	Low-BW Sim
Resolution	Resolved

2. Test Case Reference

Test Case ID	TC-0473
Title	Test that background sync correctly handles invalid input when the user revokes ...
Execution ID	EXEC-00745
Test Run	RUN-005 — Regression Cycle 5 — Build 5.1.6
Requirement	REQ-0410 — Compatibility — Data Synchronization

3. Expected vs Observed Behaviour

Expected Behaviour	The system should complete biometric verification within the defined SLA.
Observed Behaviour	The system exhibited timeout when the battery level drops below 10%.

4. Error Details

Error Message:

```
Operation timed out after 19544ms; threshold is 9107ms.
```

Stack Trace / Log Excerpt:

```
com.samsung.browser.BluetoothModule.execute(line 379)
```

Log Reference	LOG-13021

5. Executed Steps at Time of Failure

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1	Initiate Bluetooth reconnection by pressing the 'Connect' button.	The fallback UI is displayed with a human-readable error message.
2	Enter valid credentials: username='testuser_033', password='[REDA...	The UI transitions to the next screen within 3 seconds.
3	Launch the VoiceAssistant application on the test device.	The operation completes without error.
4	Initiate configuration reload by pressing the 'Search' button.	A success toast message is displayed.
5	Record response time using the built-in performance monitor.	The audit log records the event with correct metadata.
6	Enter valid credentials: username='testuser_024', password='[REDA...	The UI transitions to the next screen within 3 seconds.

6. Root Cause Analysis

Encryption key rotation not propagated to all microservice replicas before TTL expiry.

7. Linked Defect

Defect ID	BUG-0343
Title	Timeout in Sensor Manager: Operation timed out after 19544ms; threshold is 9107m...
Severity	Trivial
Priority	Medium
Status	IN_PROGRESS
Root Cause	Encryption key rotation not propagated to all microservice replicas before TTL e...
Resolution	—
Assigned To	ENG-081

8. Historical Similar Failures

HIST-0163	FAIL-0164	Timeout	6.2.4	2021-12-05	5	Resolved
HIST-0514	FAIL-0515	Timeout	6.1.9	2024-03-06	14	Open
HIST-0748	FAIL-0749	Timeout	5.3.7	2023-10-17	3	Open

9. Synthetic Data Notice

This document is entirely synthetic and intended for AI/RAG research only.

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